

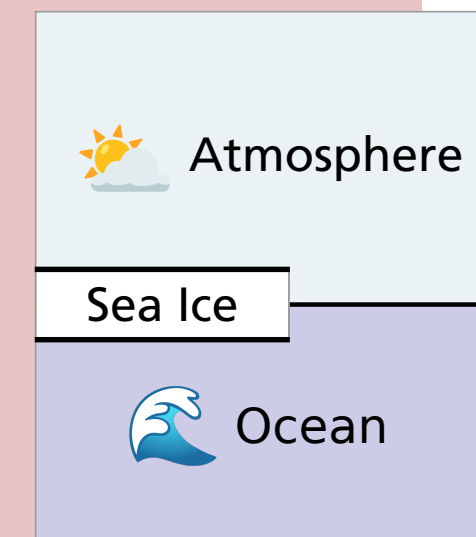
Quantifying Coupling Errors in Atmosphere-Ice-Ocean Coupling

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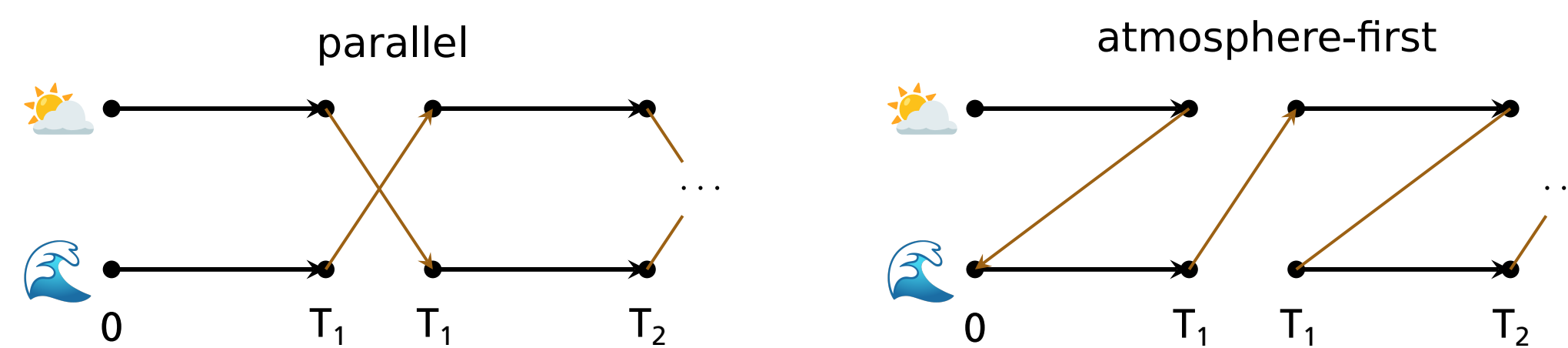
1 Introduction

In **Earth system models**, the atmosphere, ocean, and sea ice are coupled at the sea surface using boundary conditions. In other words, they solve a **coupled problem**. We use an **iterative coupling algorithm** to study numerical coupling errors in the **EC-Earth** single column model (AOSCM).

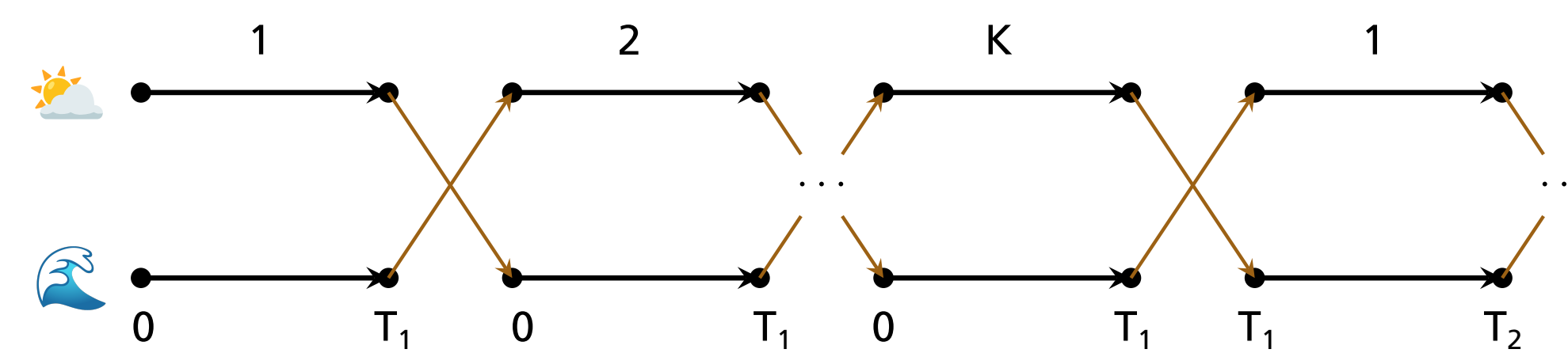


2 Method: Schwarz Waveform Relaxation (SWR)

A simulation is divided into **coupling windows**. After each coupling window, the models exchange time-averaged data.



We can interpret this as **one step** of Schwarz Waveform Relaxation, an **iterative** algorithm. Not iterating is computationally cheap but leads to a numerical coupling error.

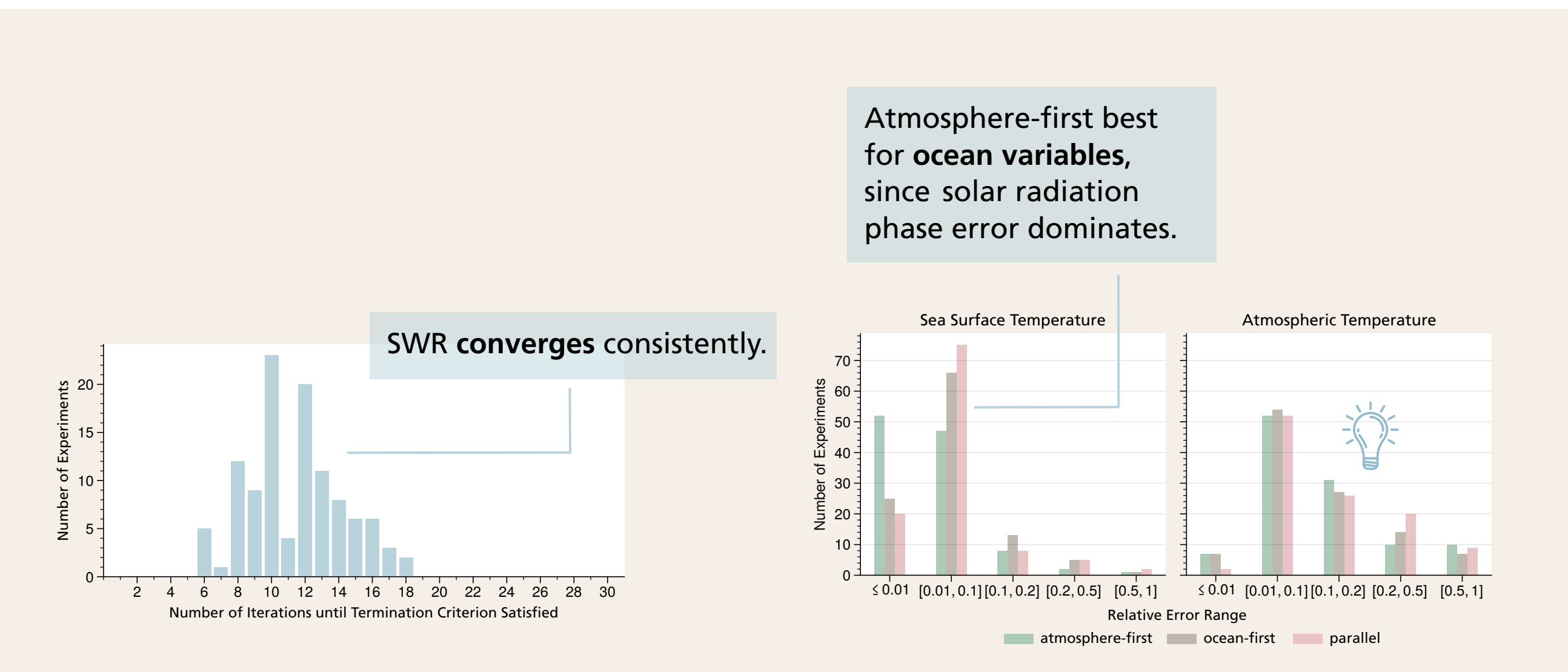


The iteration will converge if the coupled problem is set up correctly. We check whether it converges in practice and if it does, we evaluate the coupling error:

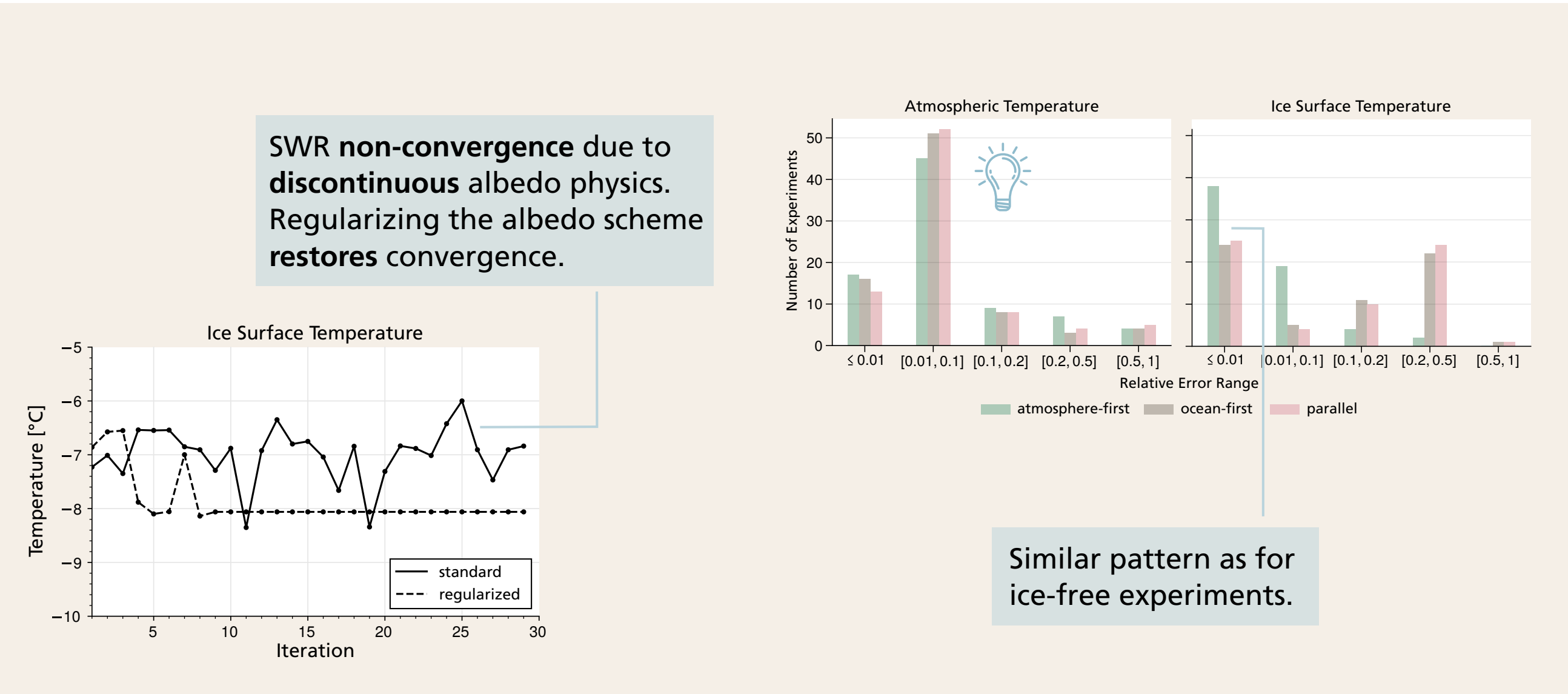
$$e_{\text{atm-first}} = \|u_{\text{atm-first}} - u_K\|$$

3 Case Study: EC-Earth AOSCM [1]

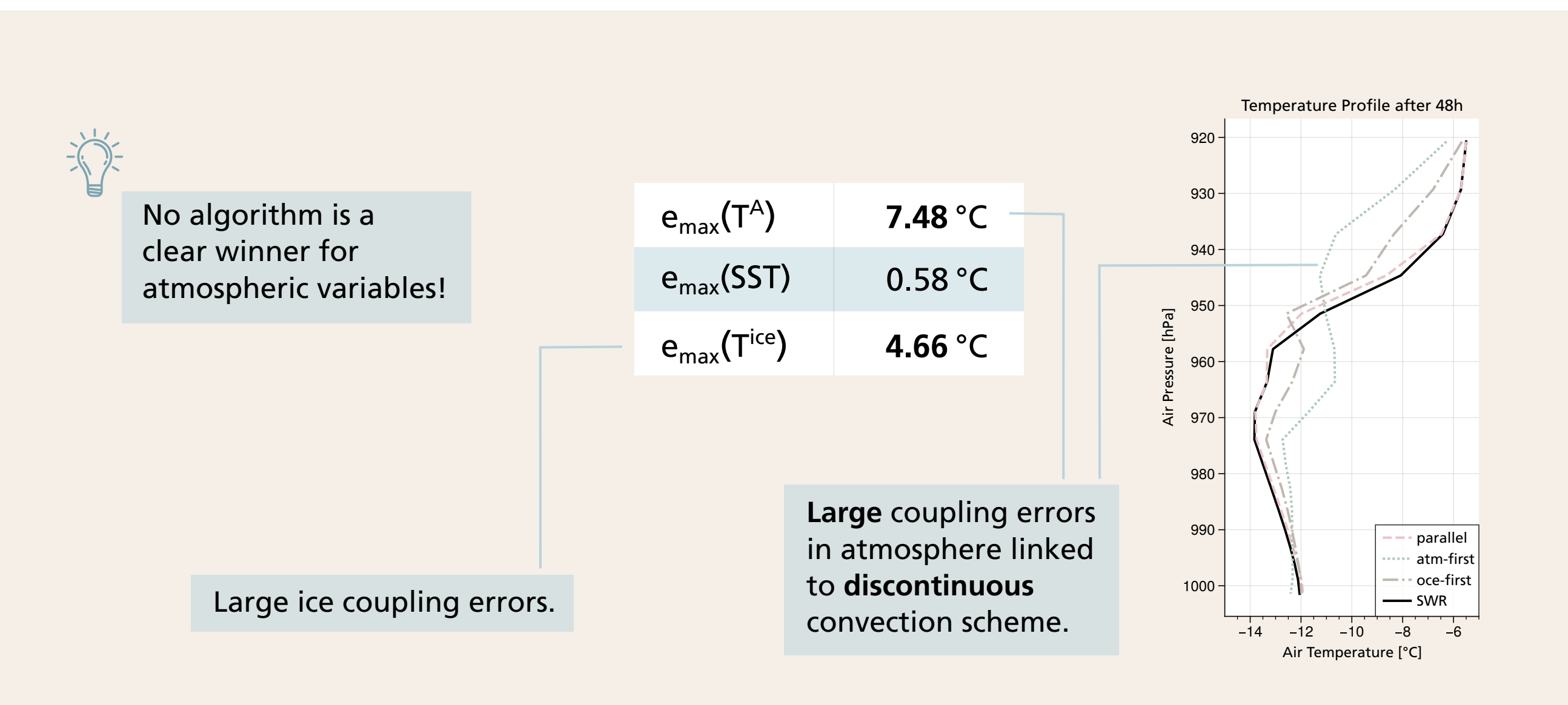
Ice-free conditions (PAPA station, July 2014, 112 two-day forecasts)



Ice-covered sea surface (MOSAIC, April 2020, 84 two-day forecasts)

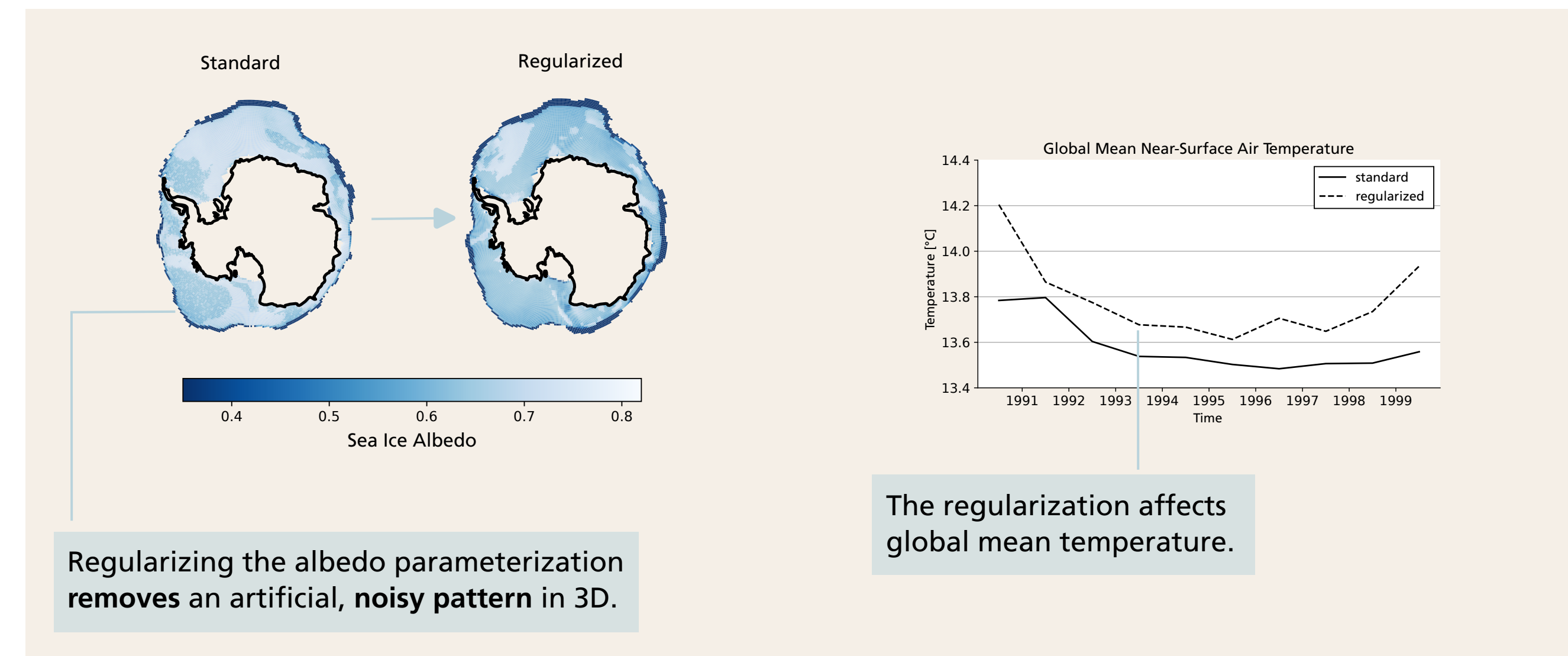


For all experiments

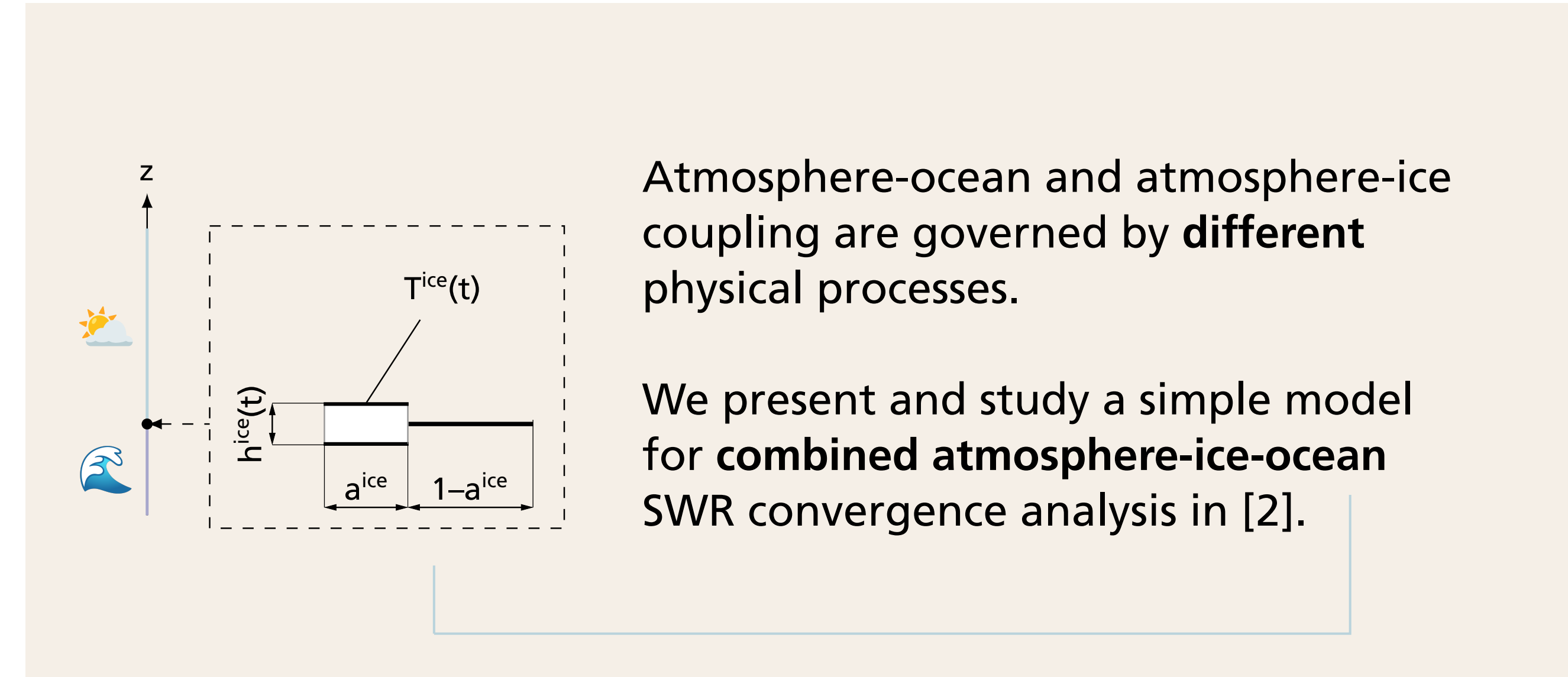


4 Learning from the Case Study

EC-Earth 4: Sensitivity experiment with regularized albedo



Atmosphere-ice analysis



Conclusions & References

SWR is a **tool** for model **development**:
→ Numerical coupling errors can be large.
→ Identify issues in physics parameterizations.

SWR is a **tool** for model **understanding**:
→ Which processes govern convergence of the iteration?
→ How does this affect the coupling error?

- [1] V. Schüller et al. (2025): Quantifying coupling errors in atmosphere-ocean-sea ice models, Geosci. Model Dev.
- [2] V. Schüller, P. Birken, H. Kjellson (in press). Analysis of Bulk Interface Conditions for Atmosphere-Ice-Ocean Coupling: DD29 Proceedings.